

UPI Fraud Analytics Dashboard

Professional Data Analyst Project Documentation

Tools: Python • MySQL/SQL • Power BI • DAX

1. Project Overview

This project analyzes UPI transaction data to identify fraud patterns, transaction risks and behavioral indicators associated with fraudulent activity. Python was used for data preparation, MySQL/SQL for structured analysis, and Power BI/DAX for interactive reporting.

2. Business Problem

The objective is to understand where and when fraudulent transactions occur and identify factors that can help prioritize fraud monitoring. The analysis focuses on risk level, banks, merchant categories, transaction types, failed attempts, risk scores and monthly trends.

3. Dataset Overview

Metric	Value
Transactions	10,000
Fraudulent transactions	965
Overall fraud rate	9.65%
Total transaction value	50.04M
Current period	2024, January–July

4. Data Preparation

- Loaded and validated the transaction dataset.
- Cleaned and standardized fields and data types.
- Created Year, Month and MonthName fields.
- Prepared a cleaned CSV for MySQL analysis.
- Validated 10,000 final records.

5. SQL Analysis

- Total transactions and legitimate/fraudulent transactions.
- Fraud rate and total transaction value.
- Fraud by risk level and risk score.
- Fraud across banks and merchant categories.
- Fraud by transaction type.
- New-device, unusual-location and unusual-amount analysis.
- Fraud rate by failed-attempt category.

6. Power BI Dashboard

- KPI cards: Total Transactions, Fraud Transactions, Fraud Rate, Total Transaction Value and Legitimate Transactions.
- Visuals: Risk Level, Bank, Merchant Category, Monthly Fraud, Monthly Fraud Rate, Transaction Type, Risk Score and Failed Attempts.
- MonthName slicer for interactive monthly analysis.

7. Important DAX Measures

Total Transactions

```
Total Transactions = COUNTROWS('upi_fraud transactions')
```

Fraud Transactions

```
Fraud Transactions = CALCULATE(COUNTROWS('upi_fraud transactions'), 'upi_fraud transactions'[FraudFlag] = 1)
```

Fraud Rate

```
Fraud Rate = DIVIDE([Fraud Transactions], [Total Transactions], 0)
```

Legitimate Transactions

```
Legitimate Transactions = [Total Transactions] - [Fraud Transactions]
```

8. Key Findings

- 965 of 10,000 transactions are fraudulent, giving an overall fraud rate of 9.65%.
- Fraud patterns vary across risk levels, banks and merchant categories.
- Monthly fraud activity changes across January–July 2024.
- Risk Score and failed-attempt analysis add further risk context.
- Multiple behavioral indicators should be considered together.

9. Business Recommendations

- Prioritize monitoring of high-risk transactions.
- Monitor banks and merchant categories with elevated fraud rates.
- Combine new-device, unusual-location and unusual-amount indicators.
- Track fraud rate over time, not only fraud counts.
- Use risk score and failed attempts as supporting signals.

10. Skills Demonstrated

Python/Pandas, data cleaning, SQL, MySQL, Power BI, DAX, data visualization, KPI development and business analysis.

11. Future Improvements

- Create a dedicated Date table and stronger star-schema model.
- Add month-over-month analysis and contribution metrics.
- Add drill-through, tooltips and dynamic titles.
- Add advanced DAX and automated refresh.

- Extend analysis to future years.

12. Conclusion

This project demonstrates an end-to-end Data Analyst workflow: Python data preparation, SQL/MySQL analysis and interactive Power BI reporting.